

Brendan Klem

P: +1 (919) 995-2110 | bklem@charlotte.edu | brendanklem.com | linkedin.com/in/brendan-klem

EDUCATION

University of North Carolina at Charlotte, Bachelor of Science in Computer Engineering Charlotte, NC
Cumulative GPA: 3.6/4.0 Graduating May 2028
Relevant Coursework: Embedded Systems, Signals and Systems, Fundamentals of Electronics and Semiconductors

TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB

3D Printing: Onshape, BambuLab, SolidWorks, CAD

Embedded & Hardware: STM32, ESP32, Arduino, GPIO, UART/Serial Communication, Sensors, BLE, Soldering, Hardware Debugging

Tools: STM32CubeIDE, Arduino IDE, Git/GitHub

Office Tools: Microsoft Office, Google Workspace, Microsoft Windows, MacOS

RELEVANT EXPERIENCE

GRACE INTELLIGENCE

Apex, NC

Internship

May 2026 – August 2026

- Independently developed a compact ESP32-S3 wearable prototype with pressure, vibration, and IMU sensors.
- Built embedded C/C++ firmware for real-time sensor acquisition, visualization, and BLE communication.
- Designed, soldered, and tested sensor hardware to develop a compact system for user data collection.
- Used Python to receive sensor data over BLE and automatically log measurements to CSV files for analysis.
- Built a custom macOS application to track and visualize real-time sensor data received wirelessly over Bluetooth.

3D PRINTING BUSINESS

Apex, NC

Entrepreneur

August 2023 – Present

- Designed custom 3D models using Onshape CAD software. Leveraged 3D printing slicers such as BambuStudio.
- Created custom designs and branded merchandise for customers.
- Worked with Green Level HS to develop and market custom school products. Built an Instagram marketing plan.
- Designed functional replacement parts and custom components to repair or improve existing products.

OTHER EXPERIENCE

TOWN & COUNTRY HARDWARE

Cary, NC

Sales Associate

May 2025 – August 2025

- Assembled and registered power tools, furniture, and grills for store display and customers.
- Managed Point of Sale operations and inventory, copied keys and mixed paint for customers.
- Provided exceptional customer service by greeting and assisting customers.

ENGINEERING PROJECTS

- **Combat Robot Club:** Led a 4-person team to design, build, and program a BattleBots-style robot using Fusion 360 and Onshape.
- **49ers Racing Club:** Designed and manufactured vehicle components and trackside equipment using SolidWorks, welding, and machining.
- **Schedule Planner:** Developed an AI-assisted tool to generate .ics files for recurring calendar events.
- **Plant Monitor:** Designed and built a multi-sensor device using an Arduino microcontroller to monitor plant conditions.

EXTRACURRICULARS/VOLUNTEERING

Engineering: IEEE, Charlotte AI Research, 49ers Racing, Combat Robot Club

Fraternity: IEEE-Eta Kappa Nu, Alpha Tau Omega

Young Men's Service League: Board Member & Communications; community service and philanthropy

Kramden Institute: Computer Technician; refurbished computers for community use

Cary Jaycees: Community volunteer